



DataHER
AFRICA

Healthcare Insights Challenge

AUGUST EDITION

Participant Project Guide

Project Scenario

*You have just joined **TediCare General Hospital** as a Data & AI Analyst.*

The hospital management has noticed increasing patient complaints about long waiting times. Some patients leave before being seen, doctors are overwhelmed on certain days, and hospital resources don't seem to be allocated efficiently.

The Chief Executive Officer has asked the Data & AI team to investigate what is happening and recommend solutions. Your role is to use data to answer the hospital's questions and present recommendations to the executive board.

Learning Objectives

By the end of the project, participants will learn how to:

- Understand a business problem before analysing data
- Clean messy real-world datasets
- Design a relational data model
- Query data using SQL
- Analyse data with Python
- Build executive dashboards in Power BI
- Use AI responsibly to support analysis
- Present findings like a consultant

Dataset & Tools

Files Provided (Dataset)

patients.csv
appointments.csv
doctors.csv
departments.csv
treatments.csv
feedback.csv
staff_schedule.csv

Tools

Participants can choose the tools they're comfortable with, but everyone should produce the same final outcome.

Data Cleaning	Excel, Power Query, Python (Pandas)
Database	PostgreSQL, SQL Server, MySQL
Analysis	SQL, Python
Dashboard	Power BI
AI	ChatGPT, Microsoft Copilot, Claude

AI should assist — not replace — participants' thinking.

Project Deliverables

01	Data Cleaning Report
02	Cleaned Dataset
03	SQL Queries
04	Business Insights Report
05	Power BI Dashboard
06	10-minute Presentation

Suggested Timeline

Week 1	Understand the business and explore the data.
Week 2	Clean data and model relationships.
Week 3	Analyse with SQL/Python and generate insights.
Week 4	Build dashboard and present recommendations.

Month Plan

Week 1 — Understand the Business

Participants receive a stakeholder brief.

Business questions:

- Why are waiting times increasing?
- Which departments experience the longest delays?
- Which days are busiest?
- Are more doctors needed?
- Which patients wait the longest?
- Which departments receive the poorest feedback?

Deliverable:

- Business Questions document
- KPI definition
- Analysis plan

Week 2 — Data Engineering

Tasks include:

- Import data
- Explore tables
- Build relationships
- Clean data
- Document assumptions
- Create SQL views if needed

Deliverable:

- Clean dataset
- Data quality report

Week 3 — Analysis

Participants answer questions such as:

Operational • Average waiting time • Peak hospital hours • Average consultation duration • Missed appointments • Cancelled appointments • Doctor utilisation • Department workload	Patient Experience • Average satisfaction score • Patients most likely to complain • Common complaint themes • Repeat visits
Financial • Revenue by department • Treatment costs • Insurance coverage • Resource utilisation	Predictive (Optional) • Predict waiting time • Predict no-shows • Forecast patient volume

Deliverable: Insights report

Week 4 — Dashboard & Presentation

Participants create an Executive Dashboard including:

- Executive Summary
- Operations Page
- Patient Page
- Resources Page
- AI Page

Participants use AI to:

- Summarise insights
- Draft recommendations
- Generate executive-friendly explanations

They should also reflect on where AI helped and where human judgement was still necessary.

Final Presentation

Each participant presents as if reporting to the hospital's executive board.

Suggested structure:

- 1 The problem
- 2 Data quality issues found
- 3 Key insights
- 4 Dashboard walkthrough
- 5 Business recommendations
- 6 Next steps

Presentation time: 10–15 minutes.

Rules & Submission

Rules

- Work individually
- AI may assist but not replace your analysis
- Document assumptions
- Tell the business story, not just the numbers

Submission

Submit your SQL scripts, Power BI file, presentation slides and executive summary. Present your findings during the showcase session.

Bonus Challenges

- ★ Build a patient no-show prediction model
- ★ Create a chatbot that answers questions about the hospital's KPIs using natural language
- ★ Perform sentiment analysis on patient feedback
- ★ Build an AI-powered report summariser
- ★ Recommend staffing levels based on historical demand

Good luck, DataHER Africa participants — we can't wait to see your insights!